Lindley has pointed out that the least-squares principle in its generalized form introduces weights involving the unknown parameters, and has no justification comparable with the Gauss–Markov theorem for classical regression. It generally provides consistent estimates. The approach has an intuitive appeal when the general error pattern can be formulated, but there is little detailed information about distributions.

A further possible line of development which has not been pursued here concerns discriminant analysis. Equation (12) has a well-known role therein, and it would appear that in the generalization, a value of β which maximizes U may lead to functions useful in linear discrimination between populations with different covariance matrices.

Much of the past work on functional relationship has appeared, often somewhat incidentally, in papers concerned more specifically with regression. In other cases it has been in papers designed to show how functional relationship differs from regression. Such work has been very necessary, but this paper represents an attempt to look at the topic in its own right; clearly a consideration of just one method of point estimation is only a start.

Acknowledgements

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Discussion on Mr Sprent's Paper

Professor D. V. Lindley: Mr Sprent, in the paper read to us tonight, proposes a generalization of least squares, and he shows us that the method works in some rather complicated problems. Rather than comment directly on the paper I would like to take this opportunity of showing why, in spite of the lack of support of a Gauss–Markov type theorem, I think his generalization works and the nature of its limitations. To do this I will use a Bayesian analysis. We have had enough philosophy about this topic, let us see how it works. I have nothing up my sleeve except a few prior distributions.

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In the limited time at my disposal it is only possible to deal with the simplest case. It seems likely that more complicated situations will not involve any new matters of principle, though I have not investigated this. Let us follow Mr Sprent and consider the two-dimensional case (his equation (4)), where

$$x_i = \xi_i + \epsilon_{xi}, \quad y_i = \beta\xi_i + \epsilon_{yi}$$

and let us further assume that ϵ_{xi}, ϵ_{yi} are independent and normally distributed with zero means and known variances σ_x^2, σ_y^2 respectively. (A Bayesian has to postulate a distribution —hence normality—but the independence is introduced merely in order to simplify the algebra, as is the assumption that the variances are known.) The likelihood for n observations (also supposed independent) is therefore proportional to

$$\left(\frac{1}{\sigma_x\,\sigma_y}\right)^n \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\frac{(x_i-\xi_i)^2}{\sigma_x^2}-\frac{1}{2}\sum_{i=1}^{n}\frac{(y_i-\beta\xi_i)^2}{\sigma_y^2}\right\}, \tag{1}$$

the unknown parameters being $\xi_1, \xi_2, \ldots, \xi_n$ and β. For the moment let us assume the prior distributions of these parameters are independent and sensibly constant over the effective range of the likelihood. This is the usual assumption made in more familiar situations. Then (1) is also proportional to the joint posterior distribution of the $n+1$ parameters and the distribution of β can be obtained by integrating (1) with respect to each of the ξ_i. This is easily done by completing the square in each ξ_i, with the result that

$$p(\beta \mid \mathbf{x}, \mathbf{y}) \propto \left(\frac{1}{\sigma_y^2+\beta^2\,\sigma_x^2}\right)^{\frac{1}{2}n} \exp\left\{-\frac{1}{2}\,\frac{s_{yy}-2\beta s_{xy}+\beta^2 s_{xx}}{\sigma_y^2+\beta^2\,\sigma_x^2}\right\}, \tag{2}$$

where $s_{yy} = \Sigma y_i^2$, etc.

The expression in the exponential is $-\frac{1}{2}U$, where U is defined in Sprent's equation (5), remembering that $\sigma_{xy} = 0$. Consequently, if we ignore the factor outside the exponential the posterior distribution has its maximum at the generalized least squares estimate of the author's. Denoting this estimate by $\hat{\beta}$, U may conveniently be written

$$U = \lambda + (s_{xx}-\lambda\sigma_x^2)\,\frac{(\beta-\hat{\beta})^2}{\sigma_y^2+\beta^2\,\sigma_x^2}, \tag{3}$$

where λ is the minimum value of U (see the author's equation (6)). Hence

$$p(\beta \mid \mathbf{x}, \mathbf{y}) \propto \left(\frac{1}{\sigma_y^2+\beta^2\,\sigma_x^2}\right)^{\frac{1}{2}n} \exp\left\{-\tfrac{1}{2}(s_{xx}-\lambda\sigma_x^2)\,\frac{(\beta-\hat{\beta})^2}{\sigma_y^2+\beta^2\,\sigma_x^2}\right\}. \tag{4}$$

Now the factor outside cannot be ignored; nevertheless it is clear that for large n, its contribution will be less important than that inside the exponential. (It is easy to show that $(s_{xx}-\lambda\sigma_x^2)$ is of order n, with probability one.) In large samples we may therefore go further than Mr Sprent and remark that β is approximately normal with mean $\hat{\beta}$ and variance approximately $(\sigma_y^2+\hat{\beta}^2\,\sigma_x^2)/(s_{xx}-\lambda\sigma_x^2)$. This ignores a certain skewness in the posterior distribution due to the term $\sigma_y^2+\beta^2\,\sigma_x^2$ in the denominator. A criticism of the paper is that it has no information about the variability of the generalized least-squares estimates. It is surely bad statistical practice to give an estimate without a standard error at least, unless the problem is genuinely a point-estimation decision situation.

For the above approximation to work n would have to be rather large. Let us therefore look at (4) more closely. It is the product of an exponential factor with its maximum near $\hat{\beta}$ and another factor with a maximum at $\beta = 0$. Why does this second factor appear, and why does it favour $\beta = 0$? The answer is not hard to find. It arises from the assumed prior distribution of ξ_i. There are only two observations (x_i and y_i) that provide information about ξ_i, however large n is, and therefore the effect of the prior does not diminish with n. We therefore ought to think a little more about the distribution. We have taken a smooth distribution independent of β. Since $\eta_i = \beta\xi_i$ this means that the prior distribution of η depends on β. The assumed distribution is asymmetric in the two axes and

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this explains the "attraction" of the posterior distribution towards $\beta = 0$. It might be more reasonable to assume that the distribution of ξ_i depends on β: for example, a large β might give rise to smaller ξ's than would a small value. One possibility is to suppose the distance of (ξ_i, η_i) from the origin is distributed independently of β. If this is smooth (as also is that of β) the joint prior for $\xi_1, \ldots, \xi_n, \beta$ is proportional to $(1+\beta^2)^{\frac{1}{2}n}$ and the posterior distribution becomes

$$p(\beta \mid \mathbf{x}, \mathbf{y}) \propto \left(\frac{1+\beta^2}{\sigma_y^2+\beta^2\sigma_x^2}\right)^{\frac{1}{2}n} \exp\left\{-\tfrac{1}{2}(s_{xx}-\lambda\sigma_x^2)\frac{(\beta-\hat{\beta})^2}{\sigma_y^2+\beta^2\sigma_x^2}\right\}. \quad (5)$$

The exponential is unaltered, but the other factor varies only slightly between σ_y^{-n} at $\beta = 0$ and σ_x^{-n} as $\beta \to \infty$. The above normality assumption is therefore more reasonable in this case.

I hope the above remarks have shed some light on Bayesian methods and on generalized least squares. I have much pleasure in proposing the vote of thanks for this thought-provoking paper.

(After reading the above, Mr Beale pointed out to me that if the values of x and y had been scaled so that $\sigma_x^2 = \sigma_y^2 = 1$ and if the distance in this new scaling were supposed independent of β, then the factor outside the exponential would have been unity and we should have the generalized least-squares situation more exactly.)

Mr M. J. R. HEALY: Mr Sprent's paper contributes notably to a problem which statisticians have found oddly difficult. Speaking historically, the twin concepts of regression and correlation have so dominated the literature that the situation involving "errors in both variates" has been regarded as exceptional in practice and advanced in theory. Mr Sprent shows how many practical situations can be handled, at least to the extent of obtaining point estimates, by an intelligent application of least-squares principles. In view of the clarity of his exposition I shall follow the custom of our Society in ranging somewhat wider than he himself has chosen to do.

Two topics will be sufficient to indicate lines along which Mr Sprent's notions might be further pursued. The first of these he has to some extent deliberately eschewed by avoiding all but a general discussion of distributional topics. I would like to add to his bibliography an important paper by J. W. Tukey (1951). In effect, the approach here is by way of a bivariate distribution of "true" values. Tukey points out that the second-degree parameters of this distribution can often be estimated from components of variance and covariance which are themselves simply obtained if the data include adequate replication. From these estimates, regression and correlation coefficients can in turn be estimated, providing among other things a basis for the psychologist's "corrections for attenuation". I have ventured to modify some of Tukey's formulae in a paper of my own (Healy, 1958). The existence of an exact linear relationship is equivalent to the collapsing into singularity of the distribution, and one is led naturally to the significance tests described by E. J. Williams (1955) and to the estimate of Mr Sprent's equation (5) using estimated dispersion parameters.

One remark must be made concerning the test for plausibility of a linear relationship and the associated confidence limits. For given data, as the significance level becomes more extreme the confidence region becomes narrower and finally vanishes altogether. We may thus be led to the odd conclusion that the data make it rather unlikely that a linear relationship exists but that if it does exist it is very accurately determined.

Finally, a comment on the plausibility of the models Mr Sprent has assumed. Although I fully realize the difficulty of getting good illustrative examples, I cannot but feel that the existence of any *exact* relationship between wood girth and growth increment is not very likely on biological grounds. He and I have crossed swords before on the subject of growth curves. While I appreciate that a model on the lines of those in Section 4 can accommodate a wide variety of growth patterns, my own inclination is to regard any complicated pattern of correlations between residuals as evidence that the growth process is not being adequately represented. I have outlined an alternative approach in a paper with Dr F. B. Leech (Leech and Healy, 1959). I suspect that, in most biological applications at least, an attack

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involving distributional ideas at the "true value" level will usually be most profitable. An important exception is the calibration problem, where it is important to know whether a new technique is measuring the same quantity as an old one. A recent example of this gave me pause when some data, which, arithmetically, fitted the exact relationship hypothesis quite well, proved graphically to need something resembling a cubic. I look forward to trying out Mr Sprent's techniques on this more complicated problem.

These remarks have perhaps caught some of the biological flavour of the paper itself. The topics discussed are potentially of interest in the special fields of most of those present and it is my privilege to second the proposal for a vote of thanks to the author.

The vote of thanks was put to the meeting and carried unanimously.

Dr V. D. BARNETT: I read Mr Sprent's paper with great interest. He has given a very welcome discussion and generalization of the least-squares approach to estimating the slope parameters in this difficult situation. My own concern is with the practical details of applying such models and I should like to make one or two comments at this level. Mr Sprent points out that he has concentrated mainly on point estimation of the β, with little attention to an assessment of the accuracy of the estimates or the adequacy of the model. As earlier speakers have pointed out, we cannot avoid these considerations in any practical work, but there would seem to be a further problem which is often of equal (or even greater) importance—i.e. the estimation of the error variances and covariances $\mathbf{\Sigma}_0$ (for example, in assessing the accuracy of observers, where each may have a different bias). Mr Sprent has considered the first two points briefly in his concluding remarks, but has throughout assumed that $\mathbf{\Sigma}_0$ is known (which in truth it never is) or that some reasonable estimate is available, for example, by replication.

My experience is that we often encounter data where there is no immediate means of obtaining such estimates. In practice, then, we often need to estimate, simultaneously, both β and $\mathbf{\Sigma}_0$, assess the accuracy of our estimates and test the adequacy of the model. The literature contains a great deal of material on this wider problem, and it would seem relevant to give a brief mention of such work in this discussion. Most of the proposed methods are *ad hoc* in nature with little consideration of properties of optimality; also they deal only with the simplest cases. But they do provide practical means of handling data where a linear functional relationship seems plausible.

Most of us are familiar with the work of Wald (1940), later modified by Professor Bartlett (1949), for the two-sample case without replication, based on dividing the data into groups and using group means. In the same situation Housner and Brennan (1948) suggested an estimator of β based on order statistics. Dorff and Gurland (1961a) proposed a class of estimators including the three just mentioned. Unfortunately, the optimal estimator within this class was somewhat impracticable since it depended on the unknown real values, ξ_i. But later for replicated observations (Dorff and Gurland, 1961b) they produced estimators based on sums of squares and products which proved more efficient than those based on groupings or order. There are, of course, many other papers which might also be mentioned in this context.

Specifically on the question of error variances there is the related work of Ehrenberg (1950) on analysis of variance with inhomogeneous error variances. Estimates and their standard errors are obtained in terms of sums of squares and products of the observations.

There is a problem which arises quite often in practice, and which falls in a sense midway between the inhomogeneous analysis of variance and linear functional models. Suppose we use p different instruments to measure the same characteristic of each of a set of q individuals. We might well expect the instrument effects to vary with the individual, and need to estimate the relative accuracies of the instruments in the face of this. A simple model for this would be

$$x_{ij} = \alpha_j + \beta_j \gamma_i + \epsilon_{ij}, \quad \left.\begin{matrix} i = 1, \ldots, q \\ j = 1, \ldots, p \end{matrix}\right\} \epsilon_{ij} \text{ independent with variances } \sigma_j^2.$$

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Absorbing the constant terms α_j we have, in Mr Sprent's notation,

$$\mathbf{x}_i = \boldsymbol{\xi}_i + \boldsymbol{\epsilon}_i \quad (i = 1, \ldots, q),$$

where the model becomes

$$\mathbf{M}\boldsymbol{\xi} = \mathbf{0},$$

with

$$\mathbf{M} = \begin{bmatrix} \beta_2 & -1 & 0 & 0 & . & . & 0 \\ \beta_3 & 0 & -1 & 0 & . & . & 0 \\ . & . & . & . & . & . & . \\ . & . & . & . & . & . & . \\ \beta_p & 0 & 0 & 0 & . & . & -1 \end{bmatrix},$$

that is, we postulate a set of *pairwise* linear functional relationships between the instruments. (Here we assume $\beta_1 = 1$.)

Here we find that the sums of squares and products of the observations yield a class of asymptotically unbiased estimators for the β_j and σ_j^2. For if

$$b_{rs} = \frac{1}{q} \sum x_{ir} x_{is} \quad (r, s = 1, \ldots, p),$$

$$E(b_{r,s}) = \beta_r \beta_s V(\boldsymbol{\xi}) + \delta(r, s)\, \sigma_r \sigma_s,$$

where

$$V(\boldsymbol{\xi}) = \frac{1}{n} \sum \xi_{i1}^2,$$

and if $p > 2$ this is sufficient to obtain

$$\hat{\beta}_r(s) = \frac{b_{rs}}{b_{1s}} \quad (s \neq r),$$

$$\hat{V} = \frac{b_{1r}\, b_{1s}}{b_{rs}} \quad (s \neq r, \neq 1)$$

and hence, from $E(b_{r,s})$, estimates of the σ_r^2.

Asymptotic S.E.'s are easily obtained and we can improve on the individual estimates by taking appropriate linear combinations. For $p = 2$ we have insufficient equations and must use some other approach. Having $p > 2$ appears to produce an effect of hidden replication, and the method has similarities to Dorff and Gurland's method for $p = 2$ with replicated observations. In practice, reasonable estimates of the parameters have been obtained. In some cases S.E.'s of the $\hat{\sigma}_r^2$ were high, but if good estimates of the β_r are obtained we can use these to transform back to Ehrenberg's model and obtain better estimates of the σ_r^2 by his methods.

Professor M. S. Bartlett: I also would like to point out that if the assumptions about the errors are left general the problem of estimating their variance–covariance matrix is very difficult; and there is much to be said for emphasizing the cases which are less complicated. Now Mr Sprent has already drawn attention to the possible relevance of multivariate analysis techniques, but it does seem important to underline this by noting that his one numerical example appears just the sort of example where this type of analysis is appropriate. One does a straight canonical analysis of the x, y variables with years, the smaller canonical root being relevant to any possible linear functional relationship. Actually, if the x and y means are plotted against years, they show curvature in opposite directions, so that a linear relationship between x and y appears unlikely. This is confirmed

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by testing the significance of the smaller root by the approximate methods I have previously proposed, the significance level reaching the $P = 0{\cdot}01$ level. It might be advisable to mention, as reference has been made to Williams (1955), that this paper contains an error in claiming that exact multivariate tests exist for testing linear functional relationship, as I pointed out at the time (Bartlett, 1957). However, the approximate methods are usually adequate, and when the relevant root is insignificant may be used to obtain approximate confidence regions for the coefficients by examining what ranges of their values maintain insignificance for the canonical root. A more complicated example with two simultaneous relations has also been discussed (Bartlett, 1948).

My only other remark concerns the existence of linear functional relations in bivariate stationary time series, where it is sometimes convenient to separate the "error" variation from both variables by means of a cross-spectral analysis.

Mr E. M. L. Beale: Could I please add my thanks to Mr Sprent for a most interesting paper, but also register a small complaint about one of his concluding remarks. He suggests that one can get confidence limits for the coefficient β by observing that U has a χ^2 distribution. Mr Healy has pointed out that this can lead to peculiar results. Now surely this is analogous to the situation in non-linear regression, discussed by Daniels (1951). It is known that the sum of squares of the residuals has a χ^2 distribution with a number of degrees of freedom equal to the number of observations. But this fact does not lead to sensible confidence regions: the size of the confidence region does not reflect the accuracy of the determination. The standard practice in linear regression problems can be expressed in terms, not of the statistic U, but of the statistic $U - U\min$, where $U\min$ is the value of U obtained by using the best-fitting parameter values with the observed data. In linear regression problems this statistic has a χ^2 distribution with p degrees of freedom, where p is the number of parameters being estimated. In non-linear regression one can get a reasonable approximate confidence region by pretending that this has a χ^2 distribution with p degrees of freedom. The properties of such regions are discussed by Beale (1960). I imagine that they would have similar properties in the more general context considered by Mr Sprent.

Dr S. C. Pearce: I very much welcome a paper on this subject, because in the past many people, mostly non-statisticians, have used regression techniques in the belief that they were thereby establishing natural laws. When statisticians have pointed out that they were doing nothing of the sort, the experimenters have reasonably asked what techniques they ought to have been using, and here there has sometimes been a quick change of subject. There have been suggestions this evening that Mr Sprent has not taken some aspects of his work far enough. Perhaps he has not, but it is nevertheless a good thing that he has begun to unify a subject of great importance that has in the past received too little systematic study.

My one comment is so elementary that it must be either trivial or fundamental. It relates to the role of σ_{xy} in equations (6), (7) and (8). From (8) it is clear that if $\sigma_{xy} = 0$, the true functional relationship must lie between the regression lines, but the more general form in (7) shows that this is not necessarily so if $\sigma_{xy} \neq 0$. Finding that $\sigma_{xy} \neq 0$ means that whatever causes a deviation in x is liable also to cause a deviation in y, as happens, for example, when a factor that ought to be held constant over an investigation does in fact vary somewhat, so leading to related changes in some of the variates. This could be expressed by saying that there is a functional relationship secondary to the one under study. However, since there is rarely any way of telling how these two relationships are themselves related, there is no certain way of estimating either of them. In fact, we cannot proceed unless there are grounds for believing that (6), or in the multivariate case (12), which is its analogue, has at least one root that is zero apart from the operation of independent random errors.

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Dr D. R. BRILLINGER: Among the problems raised in the last section of Mr Sprent's paper and noted by the previous speakers is that of setting confidence limits for his proposed estimates. Given the nature of these estimates, that is, that they are evidently best calculated by means of a computer and in addition no simple formula for their variance has been presented, this would appear to be an ideal place to use Professor J. W. Tukey's jack-knife procedure.

The jack-knife proceeds as follows: given $n = rs$ observations, divide them into r groups of s. Let q denote an estimate of a parameter of interest based on all n observations and let q_i denote an estimate based on the $s(r-1)$ observations obtained by omitting the ith group. Define the pseudo-values $q_{(i)} = rq-(r-1)q_i$. The estimate $\bar{q} = \Sigma q_{(i)}/r$ eliminates bias of order $1/n$ in the manner of Quenouille (1956), while, more importantly,

$$S^2 = \frac{1}{r(r-1)}\Sigma(q_{(i)}-\bar{q})^2 = \frac{r-1}{r}(\Sigma q_i^2-(\Sigma q_i)^2/r)$$

provides an estimate of the variances of q and $\bar{q}$. Confidence limits may now be set using the t distribution on $r-1$ degrees of freedom.

Professor Tukey has justified the use of the estimate S^2 for the class of sequilinear estimates. Justifications of the use of S^2 and indicating an alternate estimator run as follows: let θ denote the parameter of interest. Suppose

$$\text{(i)} \quad \text{ave}\, q = \theta + o(1/s),$$

$$\text{(ii)} \quad \text{ave}\, q_i = \theta + o(1/s),$$

$$\text{(iii)} \quad \text{var}\, q = \sigma^2/n + o(1/s),$$

$$\text{(iv)} \quad \text{var}\, q_i = \sigma^2/s(r-1) + o(1/s).$$

We see that to $o(1/s)$, $\text{ave}\, q_i^2 = \theta^2 + \sigma^2/s(r-1)$, $\text{ave}\, q^2 = \theta^2 + \sigma^2/n$ and therefore

$$\text{ave}\,\{(r-1)/r\}(\Sigma q_i^2 - rq^2) = \text{var}\, q + o(1/s).$$

Alternatively, if we have (iv) and

$$\text{(v)} \quad \text{corr}\,\{q_i, q_j\} = (r-2)/(r-1) + o(1/s) \quad (i \neq j),$$

then

$$\text{ave}\, S^2 = \sigma^2/n + o(1/s) = \text{var}\, q.$$

(The motivation for (v) is the fact that q_i and q_j have $r-2$ groups in common out of the $r-1$ available.)

Note that we have made no assumption concerning the independence of the observations. Also note that the assumptions (i) to (iv) indicate properties that we seek to achieve by the division into groups and the form of q_i. In connection with this last, (iv) may be achieved on occasion by applying a missing-values technique. That is, we have taken n observations, but s are to be looked upon as missing, leading to the estimate q_i. We have taken q_i based on $s(r-1)$ rather than s observations in the hope that the remainder terms in (ii) and (iv) are smaller.

Returning to the specific estimates of Mr Sprent's paper, by the imposition of conditions on $\boldsymbol{\Sigma}_0$ we can hope to satisfy (i) to (iv). The sort of conditions I have in mind are that the $(\epsilon_{xi}, \epsilon_{yi})$ come from a stationary process or that they have independence properties in the manner of the example in Section 2 of the paper.

In conclusion, let me note that the jack-knife procedure extends in an immediate manner to the multiple parameter situation. Two recent references devoted to the jack-knife are Miller (1964) and Brillinger (1964).

The following written contribution was received after the meeting:

Mr P. R. FISK: Mr Sprent appears to say, in Section 1, that a constraint on $\boldsymbol{\beta}$ to determine a constant of proportionality is sufficient to obtain a unique solution. This is

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clearly not so when $p>2$ because more than one vector $\boldsymbol{\beta}$ may satisfy his equation (1); that is, the points $\boldsymbol{\xi}$ may all lie on a $(p-a)$-dimensional plane, $a \geqslant 2$. A further condition is required to ensure that we are estimating a unique vector $\boldsymbol{\beta}$, namely, that the smallest root of (6) should converge uniquely to unity in probability. This assumption would need to be tested.

The complex model described in the latter part of Section 4 is not easy to interpret. Whilst it is true that $\boldsymbol{\beta}'\mathbf{x}$ is still a null-variate under the assumptions of the model, I fail to see that this is of any practical usefulness. Unlike the previous models, this model does not allow us to interpret ϵ as a measurement error. For any particular organism at time t the "true" measurements are $\xi^* = \alpha_x + (\beta_\xi + \phi_x)\,t$ and $\eta^* = \alpha_y + (\beta_\eta + \phi_y)\,t$ and the δ represent deviations from these "true" values. Using the arguments accompanying equation (14), we obtain the relation between η^* and ξ^* for a particular organism as

$$\eta^* = \alpha^* + \beta^*\,\xi^*,$$

where

$$\alpha^* = \alpha_y - \beta^*\alpha_x$$

and

$$\beta^* = (\beta_\eta + \phi_y)/(\beta_\xi + \phi_x).$$

The values of α^* and β^* will differ from organism to organism for a constant value of ξ^*. Information on a single organism is of limited usefulness. The sort of question that we are likely to ask is: "If we selected only those members of our heterogeneous population with a common value ξ^* what is the mean value of η^* for our selected group?" This question cannot be answered via the null-variate $\boldsymbol{\beta}'\mathbf{x}$. As an example, suppose the α and ϕ are distributed in our heterogeneous population as four-dimensional normal variates. The conditional mean of η^* for a given ξ^* would be

$$E(\eta^* \mid \xi^*) = \beta_\eta\, t + (\xi^* - \beta_\xi\, t)\,(k\sigma_\alpha^2 + mt^2\,\sigma_\phi^2)/(\sigma_\alpha^2 + t^2\,\sigma_\phi^2).$$

Lastly, I fail to see how $\boldsymbol{\Sigma}_{ij}$, which was defined in Section 2 as the covariance matrix between organisms i and j, can be non-null when the α, ϕ and δ are assumed independent from one organism to another. The formulae given by Mr Sprent would be valid if his method were applied to q measurements, made at q different points in time, on a single organism chosen at random from our heterogeneous population. Methods applicable to single organisms are not generally recommended by biometricians.

Mr SPRENT replied briefly at the meeting and subsequently in writing as follows:

When I was asked to read this paper I doubted whether the work was sufficiently developed, but it was put to me that presentation at this stage might stimulate useful discussion. This seems to have been the case, and I would like to thank all contributors for their constructive suggestions. The many additional references they have introduced clearly demonstrate the wide ramifications of the problem.

A dominant theme in the discussion was interval estimation. In Section 5 of my paper I pointed to the importance of confidence limits for estimates, and the discussion has revealed a number of possible approaches.

Assumptions about the pattern of $\boldsymbol{\Sigma}_0$ were mentioned by several speakers, and I agree that in many practical situations it is sensible and advantageous to make specific simplifying assumptions. Since my paper was written, Rao (1965) has discussed patterns in covariance matrices corresponding to the arbitrary matrix of his 1959 paper.

Turning to specific points, Professor Lindley's Bayesian discussion of the generalized least-squares procedure throws light on questions of justification of the method. Further investigation of the classes of prior distributions that lead either approximately or exactly to generalized least-squares estimates for maximizing the posterior distribution would be of interest. Professor Barnard (1964) remarked, "We statisticians have wandered around in Hilbert space for many years, even though we have not always been aware of it."

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After hearing Professor Lindley's comments I am wondering if my partiality for generalized least squares may not imply that I am a Bayesian without even being aware of it.

Professor Lindley's equation (3) makes it clear that a factor $(b_{xx}-\lambda\sigma_{xx})$ analogous to his $(s_{xx}-\lambda\sigma_x^2)$ has been omitted from the expression for Q_1/Q_2 in Section 1 of my paper; fortunately this does not invalidate any results.

Mr Healy has justifiably taken me to task over my numerical example, and I think the main thing to be said for it is that its implausibility demonstrates the dangers of using techniques blindly. For his calibration problem I feel I should have provided him with tests for adequacy and standard errors; the discussion has partly filled this gap. The paper by Tukey he refers to (Tukey, 1951) and his own very lucid paper (Healy, 1958) give an interesting alternative approach.

As Dr Barnett points out, $\mathbf{\Sigma}_0$ may not be immediately estimable, and methods such as those in the references he gives must then be used. Estimates based on replication are, however, available for many designed experiments on perennial crops or animals; in these, repeated measurements in time are often taken on the same units and the error line in a multivariate analysis of variance gives estimates of the elements of $\mathbf{\Sigma}_0$.

Mr Beale suggests an approximate χ^2 with p degrees of freedom for confidence limits, a suggestion I hope to follow up. Professor Bartlett has indicated that Williams's tests (Williams, 1955) are only approximate and this is a field where approximate tests used sensibly give reasonable results.

The jack-knife procedures outlined by Dr Brillinger seem very interesting; these techniques had escaped my notice.

Both Mr Healy and Dr Pearce have alluded to the past confusion about functional relationships due to the dominance in the literature of regression concepts. For small errors any method gives rather similar point estimates and when only these are required in the bivariate case the most sensible practical procedure may be to draw a graph; however, in biological contexts (and I suspect in other fields such as economics) there may be considerable variation about the linear relationship, and it is then that different patterns become important.

In developing the methods of this paper I had in mind fitting a relationship representing a mean effect for a treatment applied to a number of individual organisms; having developed tests and confidence limits for this, the next step would be to compare relationships corresponding to different treatments. I did not make this clear in my paper, and this seems to have confused Mr Fisk. I owe him an apology for this, but his comments give me an opportunity to clear up the matter. His approach may be appropriate to the regression type model he introduces, but this is not the problem I had in mind. In the context of Section 4, $\mathbf{\Sigma}_{ij}$ is the matrix of covariances for departures at times t_i and t_j of individual observations from the linear relationship $\eta = \beta\xi$, the mean relationship for a number of similarly treated organisms. Thus my method is designed to avoid application to single organisms, a procedure Mr Fisk very rightly criticizes. Mr Fisk also mentions difficulties of uniqueness when $p>2$; it was to avoid these that I restricted the argument at the beginning of Section 3 to a single relationship.

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